

NEURAL RMF Individualized EEG Monitoring for Early Warnings of Epileptic Seizure Risk

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Abstract

Epileptic seizures are heterogeneous events whose clinical and electroencephalographic expression can vary across people and across episodes in the same person. This work presents NEURAL RMF, an individualized EEG monitoring system based on a Resonant Memory Field. Rather than training on a patient corpus or estimating the exact time of a seizure, the system constructs an eight-minute reference of the baseline state of a session and observes how subsequent activity departs from it. The output is communicated as a traffic-light system: green for behavior compatible with the reference, yellow for early change, orange for sustained activity under observation, and red for a high-risk trajectory. Red requires a complete temporal sequence of accumulation, peak change, collective reorganization, and persistence; it is not triggered by an isolated oscillation. Eighty trajectories from public CHB-MIT and Siena recordings were retrospectively evaluated using four bilateral temporal derivations selected in previous comparative tests against higher-channel configurations. After labels were withheld during processing, 69 annotated seizures fell outside the calibration window; 66 showed a preceding red alert. The remaining three cases were affected by the availability of a basal reference or data quality and are reported as insufficient-calibration cases, not as conclusive evidence of absent signal. Red trajectories without a clinical label were also observed; they are retained as review candidates, not as confirmed seizures or false alarms. The results support the technical feasibility of personalized, label-free monitoring to describe risk states, but prospective clinical validation, video-EEG, and expert review are required before any clinical use.

Keywords: epilepsy, EEG, individualized monitoring, early warning, resonant memory field, preictal, label-free calibration.

1 Introduction

Epilepsy monitoring faces a practical tension: a seizure can occur at an uncertain time, while the EEG signal changes continuously and not every change culminates in a clinically visible event. Machine-learning approaches to seizure detection and prediction have demonstrated the value of public data, temporal evaluation, and patient-specific analysis. However, they often depend on historical labels, large datasets, and supervised training.

NEURAL RMF begins from a different question: if the relevant reference is the person's own recent baseline state, can a system describe a risk transition without having been previously trained on that person's seizures? The proposal does not claim to predict a specific time or an inevitable seizure. Its function is to monitor changes relative to an individualized calibration and communicate an early warning when a sufficiently consistent reorganization trajectory appears.

RMF stands for Resonant Memory Field. It is an information representation and retrieval system that models what has been incorporated as a dynamic geometry on a high-dimensional spherical manifold. Unlike a conventional vector database, it does not retain patterns as discrete records and then search for the nearest one: patterns are absorbed as persistent deformations of a collective state. Retrieval occurs through resonance; the system responds with greater amplitude when a stimulus matches patterns already incorporated.

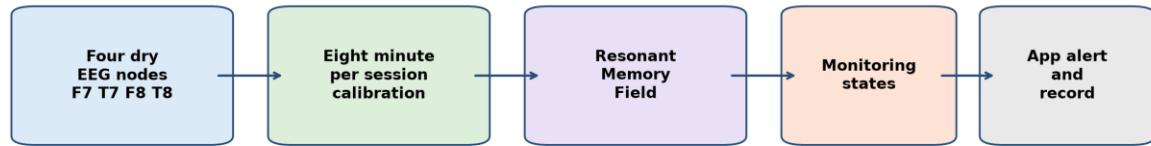
RMF exhibits three properties of interest for monitoring: continuous novelty, intrinsic temporality, and structured forgetting. Continuous novelty arises when a pattern cannot be absorbed by the incorporated geometry and produces a signal proportional to its distance from what is known, without a supervised classifier or a global fixed threshold. Intrinsic temporality preserves a different footprint for recent and older patterns without explicit temporal metadata. Structured forgetting selectively releases capacity while more stably preserving frequent or protected patterns. In this manuscript, the architecture is described as a dynamic representation framework based on spherical geometry, resonant memory, intrinsic temporality, and individualized calibration; this is not an established clinical or bibliometric category.

The four-derivation configuration was not an arbitrary choice. In the project's internal comparative tests, reducing the signal to a bilateral temporal chain retained and concentrated the information most useful to the field. With F7, T7, F8, and T8, configured as bilateral temporal derivations, the pipeline achieved better technical detection than with six derivations and with the complete 23-channel montage. This result is interpreted as technical superiority within this pipeline and these recordings, not as a claim of universal clinical superiority over diagnostic multichannel EEG.

The operational hypothesis is that the bilateral frontotemporal projection concentrates relevant focal dynamics, whereas incorporating uninvolved regions may dilute that signal during the field's dimensionality-reduction step. In an internal comparison of 19 CHB-MIT runs, the four-channel montage recorded 18 of 19 detections with a mean lead time of 67.2 minutes; six channels recorded 16 of 19, and 23 channels recorded 14 of 19. In Siena, the same four channels showed 97 percent detection and 89.5 minutes of mean lead time in the internal analysis. These figures require a reproducible protocol and independent review before being presented as definitive performance estimates.

Conventional clinical EEG and the hypothetical device serve distinct and complementary functions. Nineteen- or 23-channel EEG remains the diagnostic and clinical-validation tool, with medical context, video, and expert review. The four-node headband is proposed as a continuous individualized monitor: it seeks to record transitions relative to the user's own reference between clinical studies, not to replace hospital EEG or issue a diagnosis.

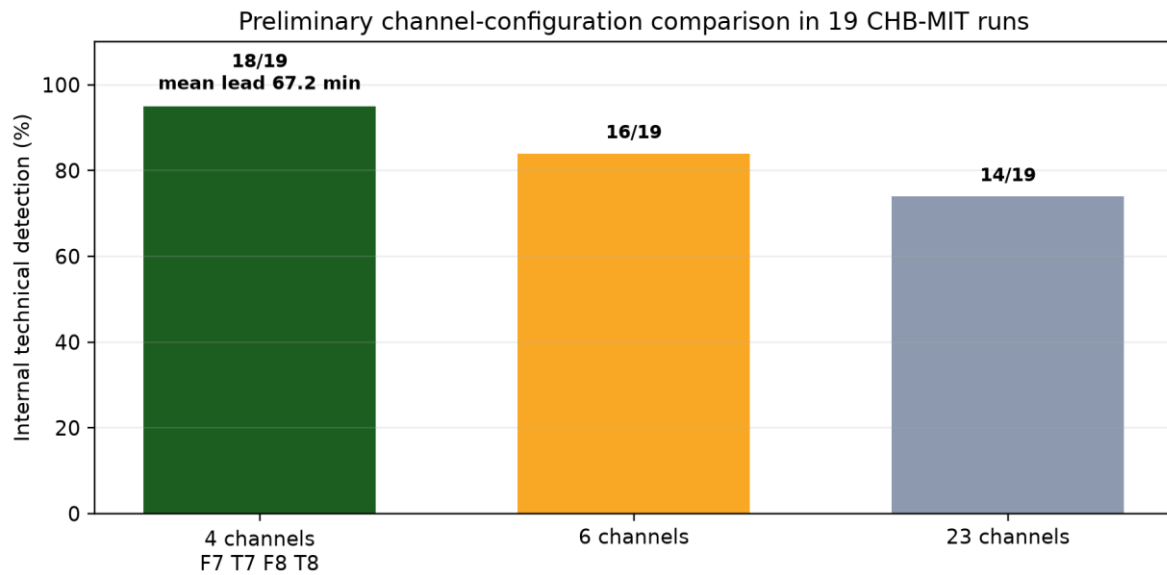
Architecture and channel selection



Clinical labels remain hidden during inference and are revealed only for retrospective comparison.

Multichannel clinical EEG and video EEG remain complementary for diagnosis and validation; the headband does not replace them.

Figure 1. NEURAL RMF research architecture. Calibration and inference do not use clinical labels.



Internal project comparison. It is not equivalent to universal clinical superiority or a prospective study.

Figure 2. Internal technical comparison of channel configurations in 19 CHB-MIT runs. The graph does not represent a prospective clinical study.

2 Objective

To retrospectively and technically evaluate whether a pipeline without population training, calibrated for eight minutes at the beginning of each session, identifies temporal risk trajectories that precede labeled seizures in public EEG recordings. A secondary objective is to describe unlabeled trajectories without assigning them a clinical diagnosis.

3 Data and case selection

Public CHB-MIT Scalp EEG Database and Siena Scalp EEG Database recordings available locally were used. The processed corpus contains 80 session trajectories from 27 unique patients: 41 recordings from

14 patients in CHB-MIT and 39 recordings from 13 patients in Siena. Therefore, trajectories do not represent independent patients; multiple recordings belong to the same person. Both datasets contain scalp EEG and seizure time annotations. Each independent file or session was treated as a new session: calibration was not transferred between sessions, and labels were not used to adjust states.

Dataset	Population and signal	Use in this analysis
CHB-MIT	Pediatric scalp EEG, continuous recordings, and annotated seizures	Individualized session trajectories using four temporal derivations
Siena	Scalp EEG from people with epilepsy, continuous recordings, and annotated seizures	External contrast within the available corpus and trajectory review

4 Methods

4.0 Resonant Memory Field and interpretable output

During calibration, baseline EEG windows are encoded and incorporated into the Resonant Memory Field as part of its reference geometry for that session. Each subsequent window is queried against that dynamic memory. If the pattern resonates with what has been incorporated, its response is compatible with the reference; if it cannot be absorbed, a continuous novelty signal appears. The library summarizes this comparison through internal variables of novelty and collective organization. For the user or clinician, those variables are translated into an interpretable temporal trajectory: relative stability, early change, sustained activity, or high-risk alert.

This order of presentation is deliberate. A person receiving an alert does not need to interpret an isolated library metric; they need to know how compatible current activity is with the reference memory, whether the change persists, and whether the trajectory meets the temporal signature that justifies raising the monitoring state. In EEG, the domain does not redefine the RMF principle: it provides a brief baseline calibration so that the monitored system itself establishes what is known and what is novel.

4.1 Individual calibration without training

The method was not trained on the analyzed data. For every session, the first eight minutes are reserved for calibration. This stage estimates reference behavior from the four derivations available in that session. Calibration is local: if the device is removed, another session begins, or insufficient usable basal signal is available, it must be repeated. Seizure labels remain outside the computation and are consulted only at the end for retrospective comparison.

4.2 Field metrics

Metric	Operational interpretation
Novelty Max	Maximum novelty observed across nodes during a window. It summarizes the most intense change relative to the individual reference.
Collective Novelty	Collective novelty. It summarizes how far the joint field behavior departs from the calibration reference.
r field	Measure of collective organization or coherence in the field. It is interpreted through its temporal evolution, not as an isolated diagnosis.
P80 baseline threshold	80th percentile of the reference metric during calibration. It acts as an individual threshold for recognizing activity above basal variability.
Seizure onset	Time of onset annotated by the dataset. It does not enter the traffic-light computation; it is overlaid later to review temporal association.

For each 30-second window, the field produces three continuous signals derived from its internal state. These signals require neither labels nor prior patient-specific adjustment: they emerge from contrasting current-window activity with the individual calibration reference stored during the first eight minutes of the session.

The first signal, Novelty Max, captures the maximum discrepancy observed in any field node during the window. It is sensitive to intense local changes and responds rapidly to abrupt perturbations. The second signal, Collective Novelty, summarizes the average displacement of the entire field relative to the reference; by integrating the response of all nodes, it is more stable and reflects gradual sustained trends over time. Both metrics are expressed in angular-dissimilarity units: values close to zero indicate activity familiar to the field, whereas higher values indicate activity farther from the calibration territory.

The third signal, r field, measures the field's internal coherence: how aligned its node oscillations are in each window. This metric does not measure novelty but organization. Its value as a clinical indicator is dynamic rather than instantaneous: a positive r field slope, meaning that the field is progressively synchronizing, forms part of the preictal signature described in the next section.

Two individual reference parameters are automatically calculated from the ninety-six calibration windows. The P80 threshold is the 80th percentile of Novelty Max values recorded during calibration; values above it indicate activity exceeding that person's typical baseline variability in that session. The B0 reference is the median of those same values and represents the central expected activity level. Both parameters are session-specific and are not transferred across sessions or people.

The following table summarizes the five quantities used in the analysis and their operational interpretation within the alert system.

4.3 Traffic-light logic

The traffic-light system does not force every deviation to green or assume that orange is an error. Green represents the state compatible with the individual baseline; yellow, early change; orange, sustained accumulation or deviation that warrants observation. Red represents a preventive high-risk alert based on a complete trajectory, not on isolated high novelty.

- A sustained orange episode must have occurred.
- The episode must reach a novelty peak above the individualized accumulation threshold.
- Novelty must decline from that peak while r field rises, consistent with collective signal reorganization.
- The signature must persist for at least three 30-second windows.

The evaluated parameters were: 8-minute calibration, 30-second windows, peak factor 2.0, three windows for orange, three for red, five windows for the r field slope, and a minimum 10 percent decline from the peak. These values are the current analysis configuration; they are not universal clinical thresholds.

5 Observational results

5.0 Preictal reorganization finding

An additional pattern was observed in the analyzed runs: before labeled events, the field's collective-organization measure tended to increase while the change signal declined from a previous maximum. This observation, internally termed Organized Calm, is the RMF candidate signature of preictal reorganization and motivates the orange-to-red transition. The finding emerged from the project's own analysis of 80 CHB-MIT and Siena records, without external review. When contrasted with the literature, Maturana and colleagues (2020) described a different biomarker termed critical slowing, based on critical-slowness indicators in long-term intracranial recordings. The two approaches do not measure exactly the same variable, but they converge on the idea that pre-seizure dynamics may contain structured and clinically useful information.

The processed corpus yielded 80 session trajectories. Sixty-nine labeled seizures were located after calibration. Three were classified as not evaluable for the preceding-alert comparison because of documented issues with available basal reference or signal quality during technical review. In the 66 evaluable events, the red signature appeared before the annotated onset: 66 of 66. Under this eligibility criterion, detection of the project's operationally defined biomarker was 100 percent. This figure is a retrospective technical measure over evaluable events, not a prospective clinical metric or a performance estimate based on independent patients.

Measure	Result	Correct interpretation
Processed trajectories	80	Retrospective monitoring units
Labeled seizures outside calibration	69	Events available for temporal comparison
Seizures with preceding red	66 of 69	Retrospective temporal association; not equivalent to prospective clinical sensitivity
Grouped non-green trajectories	171	Episodes grouped to describe shape, duration, and proximity to a label
Unlabeled trajectories	99	Candidates for review; not automatically called seizures or false alarms

Trajectories near labeled seizures tended to be longer and more sustained than unlabeled candidates. Gradual profiles, with increasing accumulation before the maximum, and abrupt profiles were observed. The coexistence of both forms supports intra- and inter-patient heterogeneity and suggests examining whether the slope, duration, and amplitude of preictal accumulation relate to clinical phenotype, including the distinction between episodes with physical manifestations and subtle events. With the available annotations, no consistent relation between profile and clinical duration or severity can be claimed; this association remains a priority hypothesis for validation with video-EEG and medical classification.

Trajectory and cohort figures

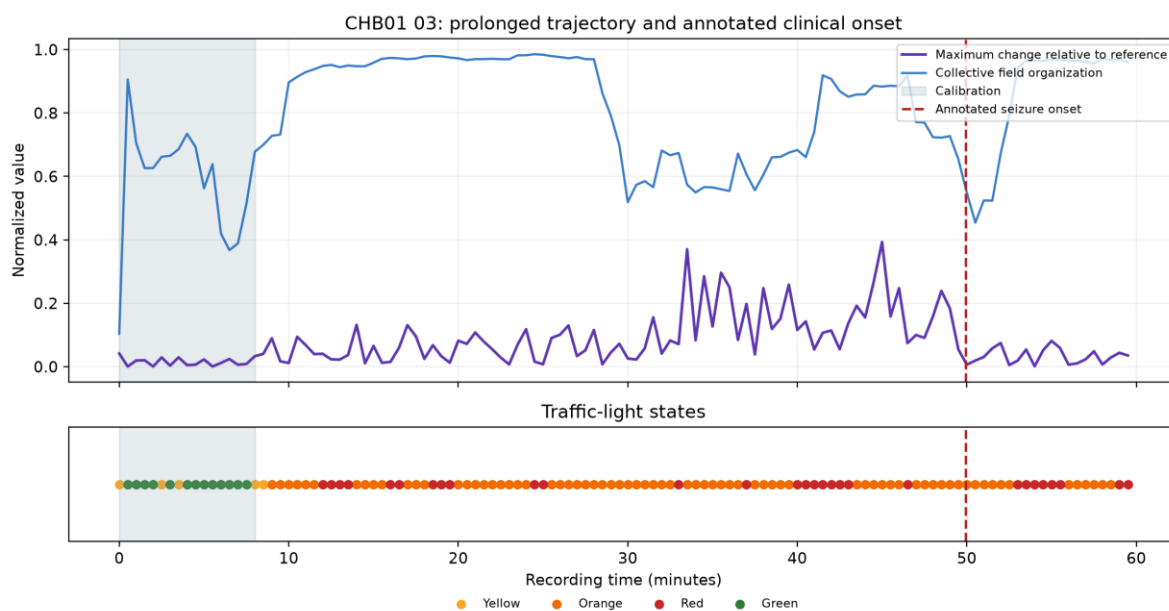


Figure 3. CHB01 03 case. The dashed red line indicates the annotated onset; the traffic-light state was produced without using that label.

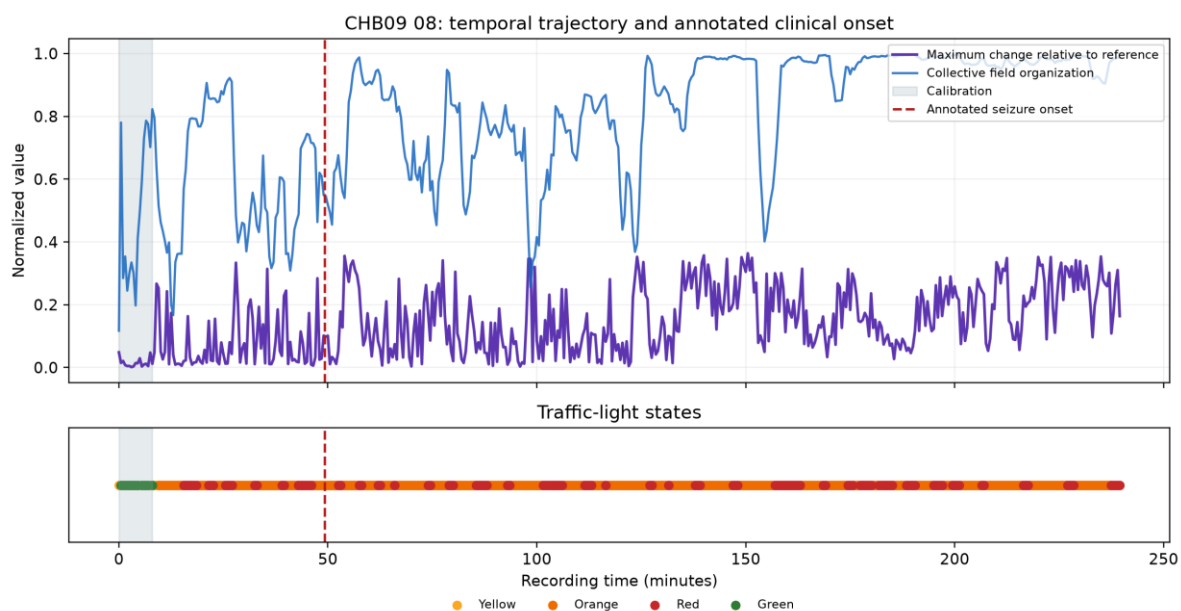


Figure 4. CHB09 08 case. Example of a temporal trajectory with dynamics different from the gradual case; no clinical severity is inferred from curve shape.

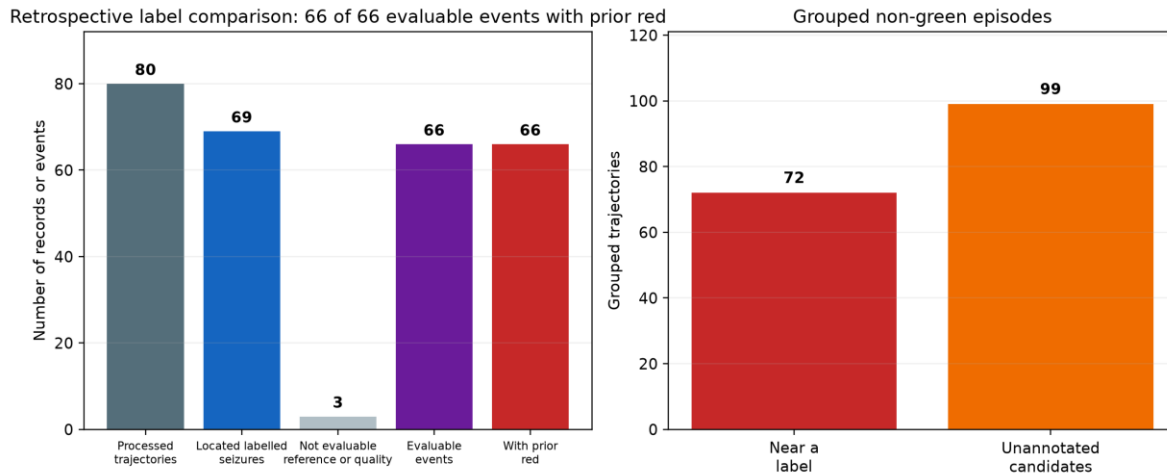
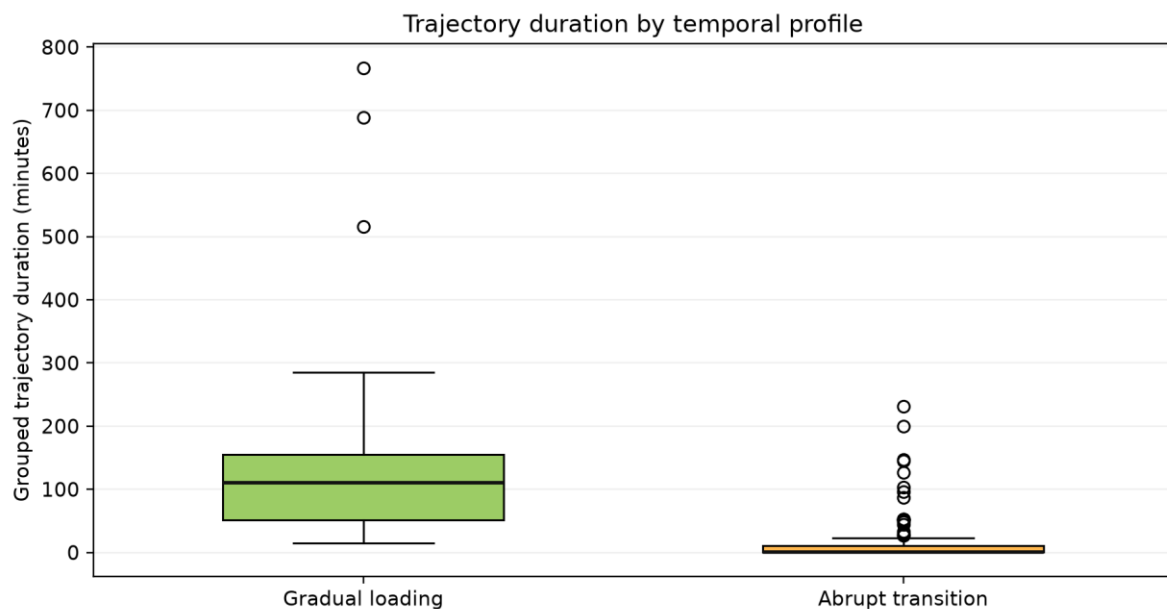


Figure 5. Summary of the retrospective comparison and grouped trajectories. Unlabeled candidates require clinical review.



Descriptive temporal profile; it does not classify seizure type or clinical severity.

Figure 6. Duration of grouped trajectories by temporal profile. Classification is descriptive and does not equal diagnosis or clinical severity.

6 Interpretation and limitations

A red trajectory without a label is not a demonstrated false alarm, but neither is it a confirmed seizure. It is a signal that the system detected a structured temporal transition relative to the individual reference: sustained activity, maximal change, reorganization, and persistence. Repetition of this form in unlabeled candidates makes the hypothesis of subtle or subclinical epileptic activity, or of an epileptic trajectory that did not culminate in a recorded clinical manifestation, plausible and clinically important to review. Transient activity, artifact, and non-epileptic physiological changes remain alternative explanations. Synchronized video-EEG, expert review, and symptom logs are needed to distinguish them.

The 66 of 69 result should be read as technical evidence of temporal association under this retrospective protocol. It does not yet demonstrate prospective performance, positive predictive value, driving safety, or clinical efficacy. At the same time, the available labels reflect events recognized and annotated by the original protocol; they are not an exhaustive description of all relevant brain activity. Tracking complete trajectories opens a research line on a continuum of activity: accumulation, alert, event, recovery, and possible subtle episodes. Alerts may remain active after a seizure because the signal has not yet returned to the basal reference. The final design must preserve that history and avoid repeated notifications within the same episode.

The duration, slope, and amplitude of the preictal trajectory are candidate variables for studying the relationship between electrophysiological evolution and clinical manifestation. The working hypothesis is that prolonged trajectories or those with marked accumulation may differ from abrupt, brief events, including subclinical candidates. This hypothesis is relevant to neurology, but it does not permit inference of clinical severity or harm from the current data.

7 Implications for a monitoring device

The proposed research architecture is a four-dry-node scalp headband at F7, T7, F8, and T8, an embedded chip that processes windows locally, and a mobile application. The application would display the traffic light, record the onset and duration of each trajectory, retain subsequent evolution, and associate every session with its own eight-minute calibration. The system would not communicate an exact seizure time; it would communicate an individualized risk transition when the red trajectory becomes established. Its intended function is complementary to conventional clinical assessment, not a replacement.

The four-node selection also responds to physical feasibility and internal technical results against six- and 23-channel configurations. The device would enable a longitudinal record of states and episodes: events with documented clinical correlates, subclinical candidates, and inconclusive transitions. A proposed Cerebral Novelty Index for the application could summarize, with traceability to the original signal, the distance of each window from the individual reference. It would not initially be a validated clinical scale; it would support review of evolution, therapeutic changes, and trends between consultations.

The same architecture could be investigated for longitudinal follow-up of neurodegenerative diseases such as Parkinson disease or Alzheimer disease. In that scenario, the purpose would not be to alert a seizure within minutes, but to quantify over weeks or months the drift of the individual's electrophysiological signature, the stability of the basal reference, and the response to therapeutic changes. A continuous personalized monitor could provide an objective complement to subjective logs and intermittent visits, helping discuss whether a treatment stabilizes, modifies, or does not change patient dynamics. This extension remains a research hypothesis: it requires specific studies, defined clinical outcomes, and validation separate from the epilepsy application.

Anti-ictal modulation is a separate research line. The accompanying simulation prototype explores whether an oscillatory intervention initiated during an early gradual trajectory could divert a model toward its interictal reference. A historical EEG record cannot demonstrate that an intervention would have prevented a seizure because it does not allow retrospective modification of a patient's brain dynamics. Therefore, any result of that simulation will be reported as a counterfactual model analysis, not as demonstrated prevention; its evaluation requires a specific preclinical and clinical protocol. In the

current historical screen, 33 gradual trajectories near labeled seizures met a minimum 25-minute window between the first red alert and annotated onset; the median was 99.3 minutes. This result establishes temporal eligibility to study the hypothesis, not the efficacy of an intervention.

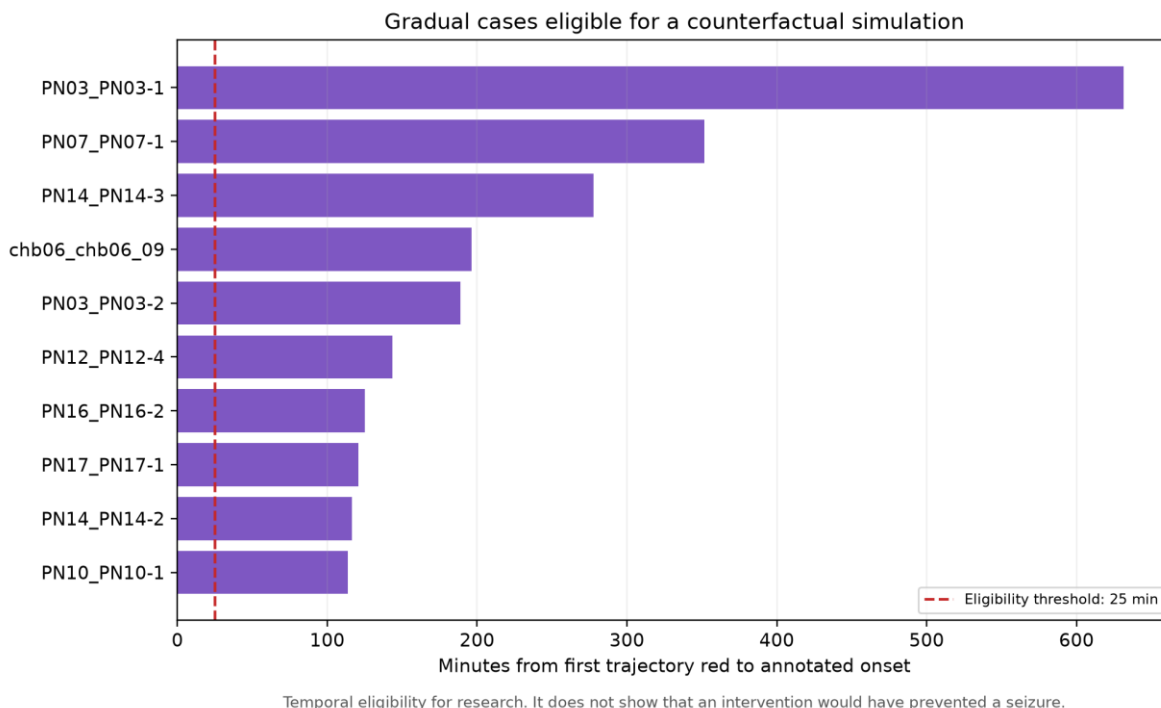


Figure 7. Ten gradual cases with a window longer than 25 minutes for counterfactual research. Temporal eligibility does not demonstrate prevention by intervention.

8 Conclusion

NEURAL RMF applies a Resonant Memory Field to individualized EEG monitoring: it constructs a session reference geometry over eight minutes and measures by resonance the compatibility of subsequent activity, without population training or labels during inference. Selection of four bilateral temporal channels was supported by internal technical comparisons against six- and 23-channel configurations. In the reviewed public data, the traffic-light trajectory logic produced preceding red alerts before most labeled seizures available for comparison and detected unlabeled candidates that justify clinical review. The finding supports a technical preprint, development of monitoring instrumentation, and a collaborative clinical-validation phase; it does not demonstrate exact prediction, seizure prevention, or medical validation of the device.

Context references

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Siena Scalp EEG Database. Scalp EEG recordings with seizure time annotations; BIDS version available on Zenodo, doi:10.5281/zenodo.10640762. Project repository: <https://github.com/Gusmal02/NEURAL-RMF>.